Stock Chart Scanner

Fundamental + Technical Analysis with Quarterly Results

What the app does

- Pick a stock universe (e.g. Nifty 500 or S&P 500).
- Pull price data and quarterly financials for each stock.
- Score every stock on **Technical** and **Fundamental** factors.
- Combine into one score and show a ranked table with a chart per stock.
- Each quarter, after results are declared, refresh fundamentals and re-rank.

Data sources (free to start)

Need	Source
Daily prices (OHLCV)	yfinance (NSE symbols end in .NS, BSE in .BO)
Quarterly results (sales, profit, EPS)	yfinance quarterly financials, or Screener.in / NSE exports
Ratios (P/E, ROE, debt/equity)	yfinance info, or Screener.in
Paid, more reliable option	Fyers, Zerodha Kite API, or Alpha Vantage

Scoring logic

Technical (price chart)	Fundamental (quarterly-based)
Trend: price above 50- and 200-day MA; golden cross (50 above 200)	**Growth:** sales growth YoY and QoQ
Momentum: RSI between 50 and 70; MACD bullish crossover	**Profit:** net profit and EPS growth YoY
Breakout: near 52-week high with volume above 20-day average	**Margins:** operating margin trend
Volatility: ATR or Bollinger Band squeeze	**Health:** ROE, debt/equity, P/E vs sector average

Each factor is scored 0-100. **Final score = 0.5 x Technical + 0.5 x Fundamental** (weights adjustable). Labels: Strong Buy, Watch, Neutral, Avoid.

Quarterly view

- Stores each stock's last 4-8 quarters of results.
- Shows QoQ and YoY growth with a **result beat / miss** flag.
- Lists stocks whose results were just declared, with the price reaction.
- Keeps score history to show which stocks improved this quarter.

Tech stack

Layer	Choice
Backend	Python (pandas, ta / pandas-ta, yfinance)
Storage	SQLite, so data is not re-fetched on every run
App	Streamlit (fastest to build, works in mobile browser) + Plotly candlestick charts
Scheduling	Daily price refresh; quarterly fundamentals refresh

Build phases

Phase	Deliverable
1. MVP	Scanner for one watchlist: technical indicators, fundamentals table, combined score
2. Charts	Candlestick with MA, RSI, MACD; filters (score, sector, market cap)
3. Quarterly module	Results history, growth trends, result-day reaction
4. Extras	Alerts (Telegram / email), backtest of the scoring rule, CSV export

Caveats

- Free data can be delayed or have gaps; verify anything you plan to act on.
- This is a screening tool, not investment advice. A high score does not guarantee returns.

Decisions needed before building

Question	Options
1. Which market?	NSE/BSE (India), US, or both
2. Which format?	Streamlit web app (recommended), mobile app, or desktop app
3. Which universe?	Nifty 50, Nifty 500, or your own watchlist